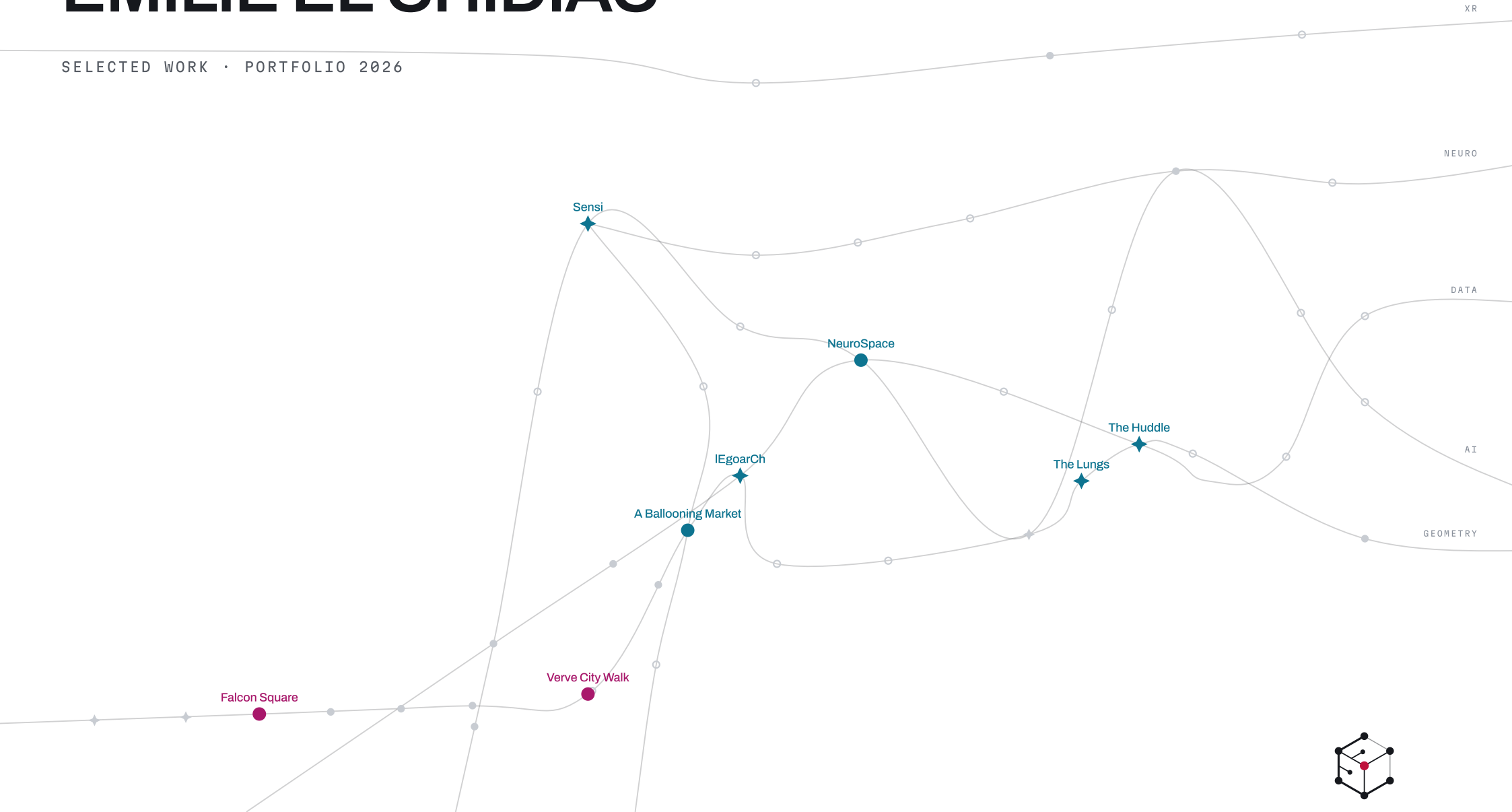


# EMILIE EL CHIDIAC

SELECTED WORK • PORTFOLIO 2026



emiliechidiac.com

*I work with design, technology and minds.*

WHO

I will not pretend I can fill a blank notebook from nothing. What I am good at is judging what the machine hands back and steering it. Someday that might be for a cursor instead of a person. Dystopian or utopian?

We build for the eye sometimes, which helps. Mostly we build for function, which is counterintuitive, isn't it? We should be building for us to function at our best, not just the building.

So I build the instruments: Sensi scores a floor plan across six senses, NeuroSpace estimates how a room reads to the brain and shows its weights so you can disagree with them.

It was never about being the most interesting person in the room. But the most interested one: what is  $x$  doing to  $y$ ?

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# Sensi

When you fix one sense in a floor plan, what happens to the other five?

TYPE computation & research  
AWARD macad awards 2026 · design copilots · winner  
TEAM macad studio · team of 4  
STACK python · langgraph · fastapi · react  
LINKS [live app](#) · [blog](#) · [github](#)

## WHAT

Every tool in the stack could tell us how a building performs. None of them would say how a room feels. Sensi closes that gap: a copilot that reads a floor plan and scores comfort across six senses (thermal, visual, acoustic, spatial, olfactory, tactile), calibrated to one person at a time, not an average. It was prototyped first as an MCP tool, then rebuilt as a standalone app. Project lead, A to Z, built by a team of four: Lakzhmy Mari Zaro, María Sánchez Domínguez, Charles Abi Chahine and me.

## WHY

Comfort is usually the thing that shows up, or does not, after the design is done. Sensi makes it a layer you can interrogate while the plan is still soft, because you do not walk into a room and average your experience: the thing that is wrong is the thing you notice. And nothing you fix stays fixed alone: the whole project is the ripple, what a change drags along.

## HOW

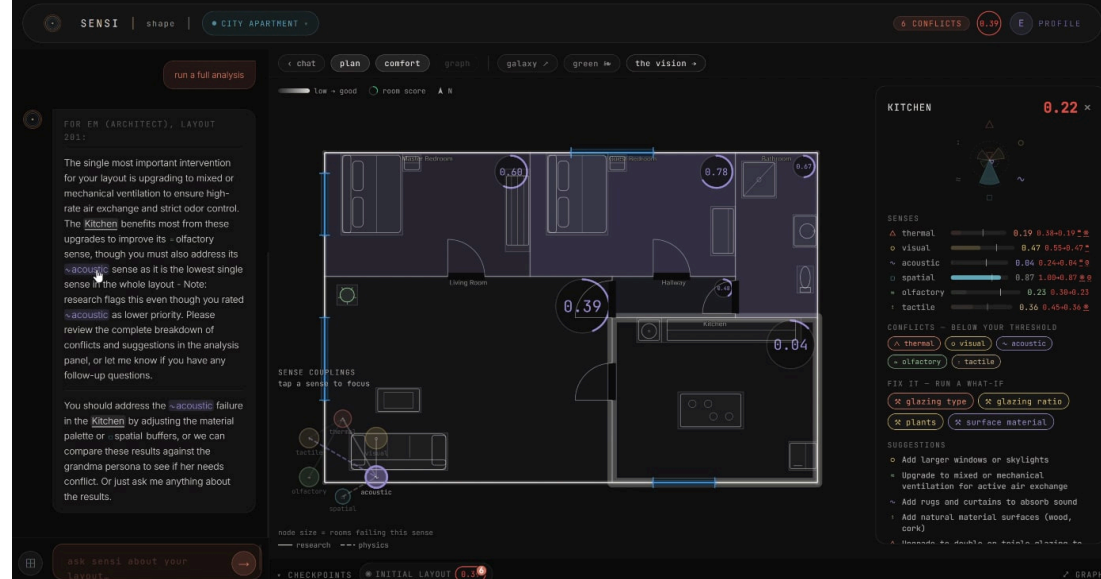
Onboarding calibrates it to one person: thermal grudges, noise tolerance.

One LLM call per turn routes each request through a LangGraph graph: analyze, edit, preview, audit. Two evals ride inside: a critic on every suggestion, an evaluator on every reply.

A coupling matrix ripples every change into the neighboring senses, so a fix that quietly breaks another score gets flagged, not hidden. *n.b. the six scores argue like a family. the coupling matrix is the dinner table.*

Edits preview before they commit, then a vision model redraws the room, structure intact, and hands over the report.

PLATE 01 · FIG 1.0



## WHAT CAME OF IT

Two LLM providers scored the same three scenes, one arranged to fail, and mostly agreed. Easy to call that validation. We wrote agreement is not truth into the notes instead, and kept every disagreement as data. It hardened into evals: swap a model and the same apartment replays through the same prompts, judged blind before the swap is believed. Nobody has stood in a room Sensi scored and said whether it was right. It estimates, it does not measure, and every coupling in the code is labeled verified or inferred, so you can see which links rest on research and which on our reasoning.

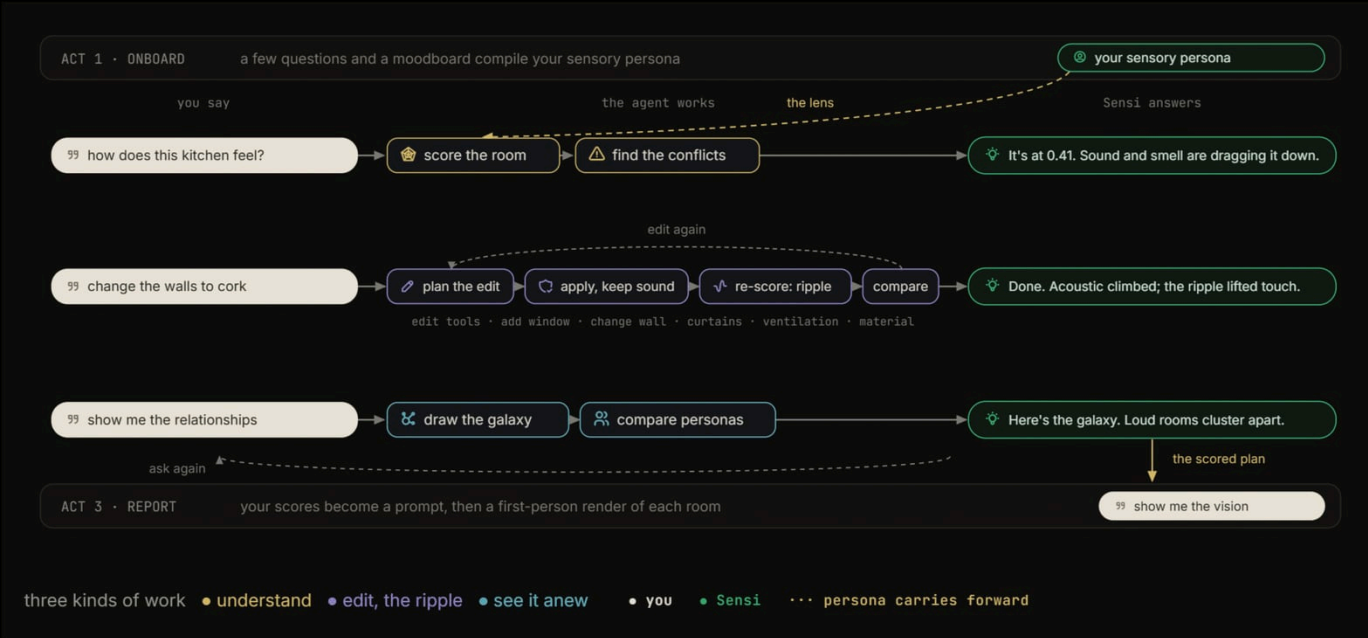


FIG 1.1 · Act 2 of the pipeline as a flowchart: score, edit, ripple and galaxy loops around the plan while it is still soft

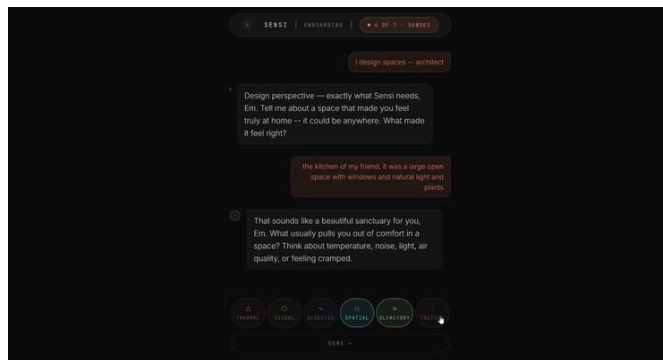


FIG 1.2 · Sensi's onboarding chat calibrating the comfort copilot to one person: their noise tolerance, their thermal grudges

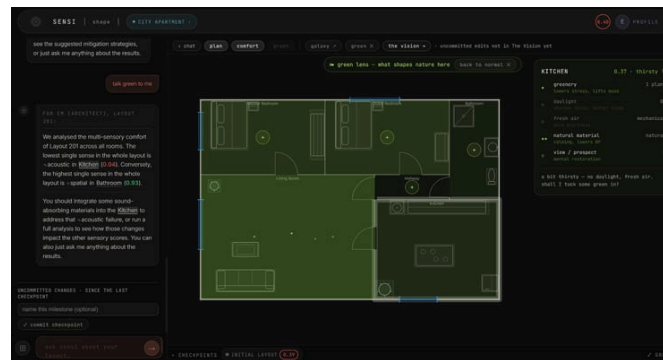


FIG 1.3 · The green lens over a city apartment plan: what shapes nature here, the kitchen flagged thirsty for daylight and plants

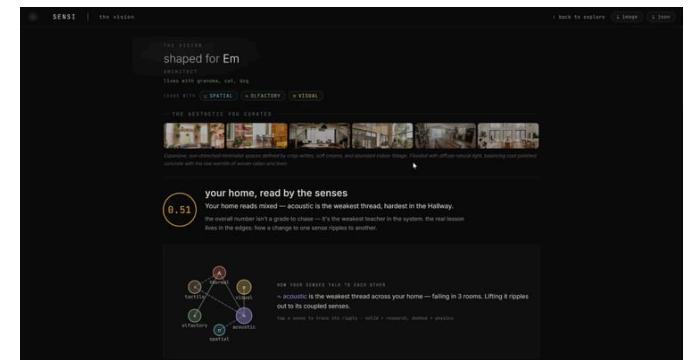


FIG 1.4 · The vision report Sensi hands over: comfort scores shaped for one person, with the edits the layout absorbed

# lEgoarCh

*How can AI turn a text prompt into a LEGO set that actually snaps together?*

TYPE computation & research

AWARD jury award

TEAM macad generative ai · with charles abi chahine

STACK flux.2 klein · trellis-2 · lora · ldraw

LINKS [github](#) · [blog](#)

## WHAT

A render is a promise, not a product: you cannot snap a JPEG together on your living-room floor. lEgoarCh takes a text prompt and returns a LEGO Architecture set that is digitally verified buildable: AI imagines it, deterministic code makes it snap together, brick by brick, out of real catalog parts. Built with Charles Abi Chahine, end to end as a pair; I stitched the stages into one flow, prompt in, set out, and gave it its interface and its story.

## WHY

The inspired gesture is cheap now; anyone can generate a thousand renders before lunch. The interesting problem moved to verification: what does it take to turn a generated image into something that provably holds together?

## HOW

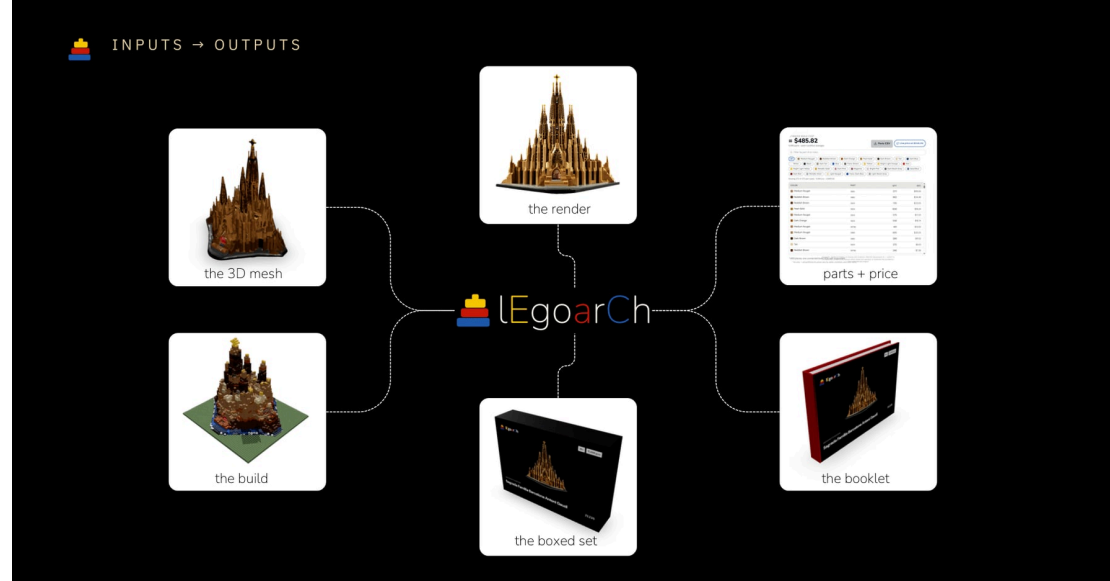
A LoRA fine-tuned on a 40-image dataset teaches FLUX the LEGO Architecture look; the prompt becomes a styled render.

TRELLIS lifts the render into a 3D mesh, and the mesh is voxelized into brick space.

An optimizer places real catalog bricks into the voxels, enforcing connectivity, support, and perceptual color accuracy.

The set exports as LDraw, the format the brick world already speaks.

PLATE 02 · FIG 2.0



## WHAT CAME OF IT

The most instructive moment was a failure: an intermediate model came back connected and supported, yet not legible as architecture. Structurally sound and visually wrong is still wrong, so legibility joined the constraints. The final pipeline was benchmarked on three buildings.

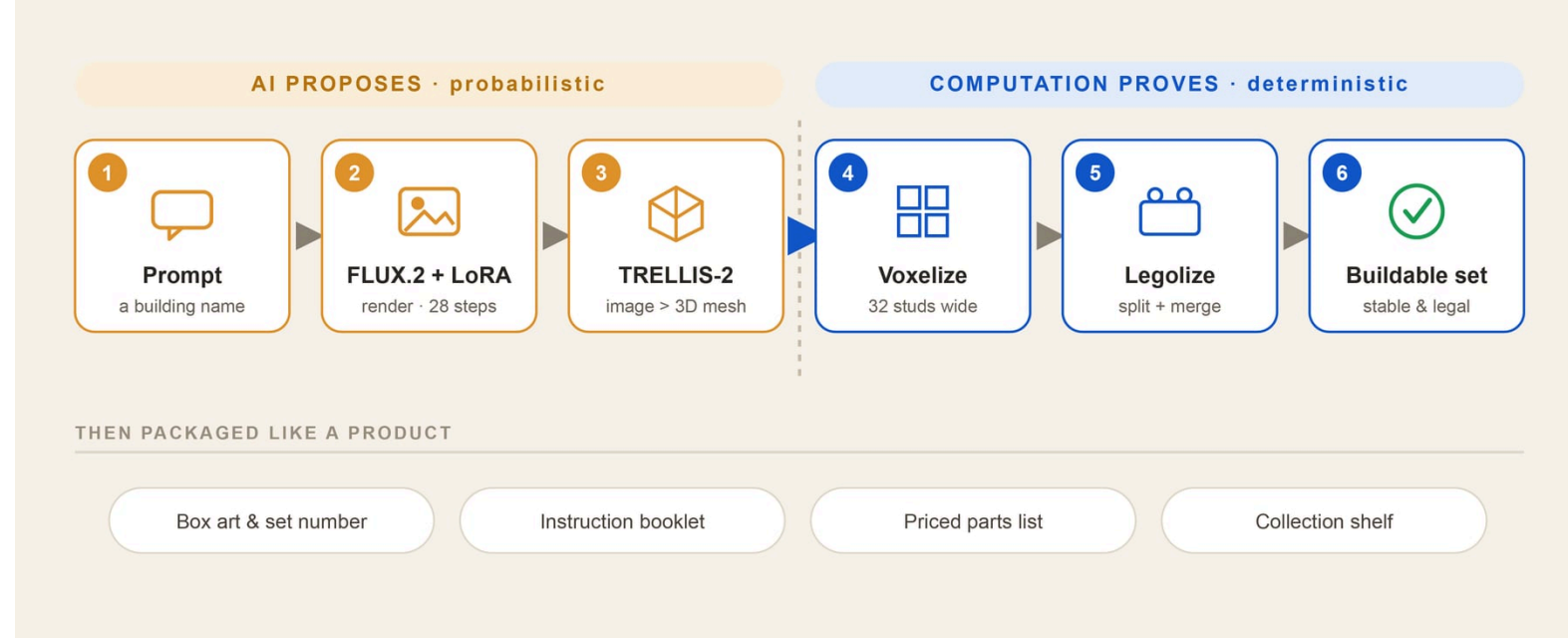


FIG 2.1 · The lEgoarch GenAI pipeline: FLUX and TRELLIS propose, deterministic code verifies the LEGO set as digitally buildable

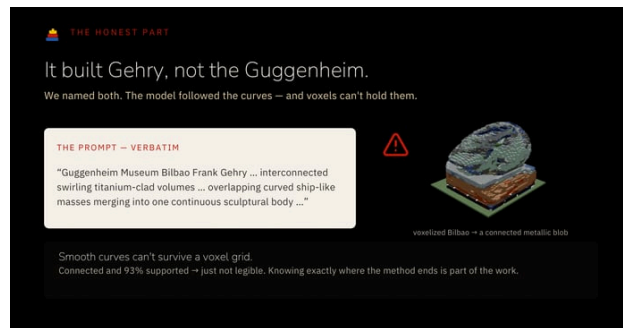


FIG 2.2 · The failure that taught the constraint: the Guggenheim came back connected and supported, and still not legible as architecture

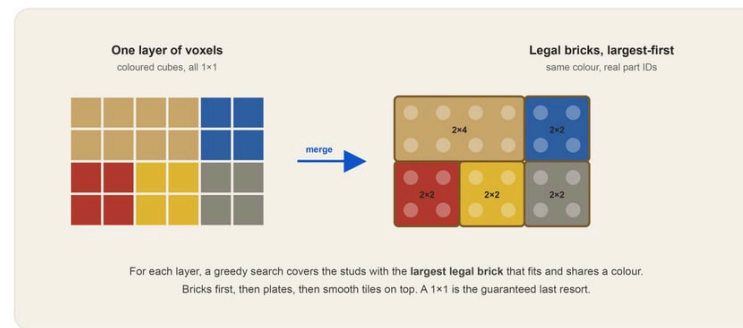


FIG 2.3 · The split and merge pass packing the voxel grid into the largest legal bricks, fewer parts holding the same shape

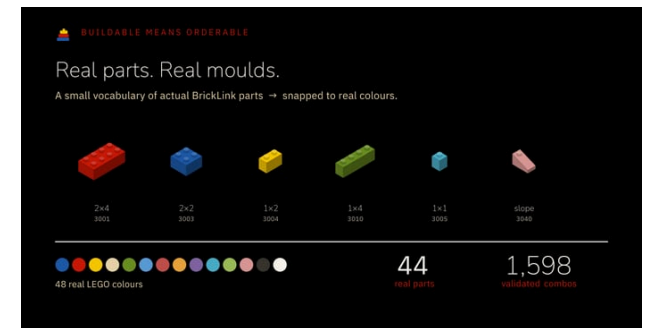


FIG 2.4 · Real parts and real molds: 44 BrickLink parts in 48 colors, the catalog the optimizer is allowed to build from



# NeuroSpace

*Can you visualize the parameters that affect how a room makes you feel?*

TYPE computation & research

TEAM macad · solo

STACK grasshopper · rhino.compute · vue 3 · three.js

LINKS [try it live, almost](#) · [no 3d](#) · [github](#) · [blog](#)

## WHAT

You are sitting in a room right now, and its defaults are quietly working on you: the ceiling height nudging your cortisol, the daylight setting your circadian clock. NeuroSpace makes that invisible layer legible: move a slider and the room rebuilds while a score answers back, live. I built it on my own: a Grasshopper definition doing the heavy geometry on the server through Rhino.Compute, Three.js drawing the room in the browser, and a scoring pass that estimates the behavioral effect the moment you let go. It is built to be played with, not read.

## WHY

This is the thesis I keep circling: BIM, reframed from Building Information Modeling to Behavior Information Modeling. The information that matters is not just what a building is made of; it is what the building is doing to the person inside it.

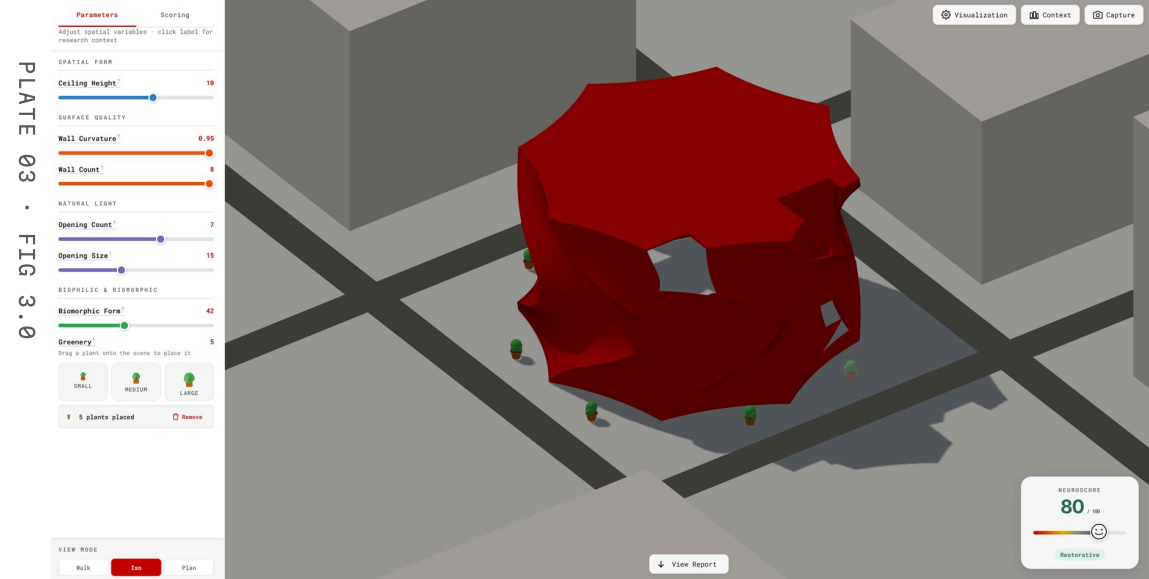
## HOW

Describe the room as parameters, not geometry: ceiling height, wall count and curvature, openings, organic form, plants. Every one is a slider.

Send the parameters to a Grasshopper definition on the server; Rhino.Compute evaluates it and streams the heavy geometry back, so the browser never has to model anything itself.

Draw the returned room with Three.js. Geometry is the slow path; it only recomputes when the shape actually changes.

Score the behavior on the fast path, in the browser, the instant a slider settles: a transparent weighted sum over the dimensions the research cares about. No server round trip, so the number answers as fast as you can drag.



## WHAT CAME OF IT

It shipped live: the app runs in the browser today, and the weights it scores with sit in the public repo, which means you can read them and argue with them. *n.b. a score you can argue with beats a number you have to trust.* The score stays a heuristic, not an instrument. It estimates; it never measures your body, and it shows every assumption on the way to the number so you are free to overrule it. When Dr. Cleo Valentine saw it, her note set the frame I kept: treat every score as a hypothesis, then watch how it translates.

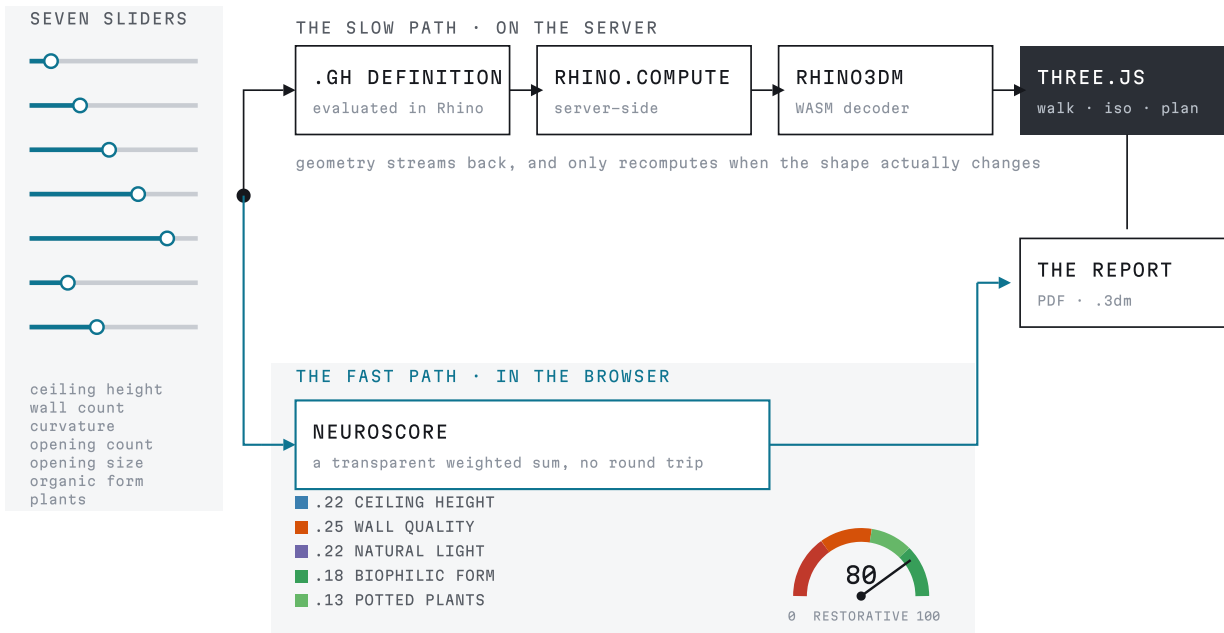


FIG 3.1 · One slider move, two paths: the server returns geometry, the browser returns a score before you let go

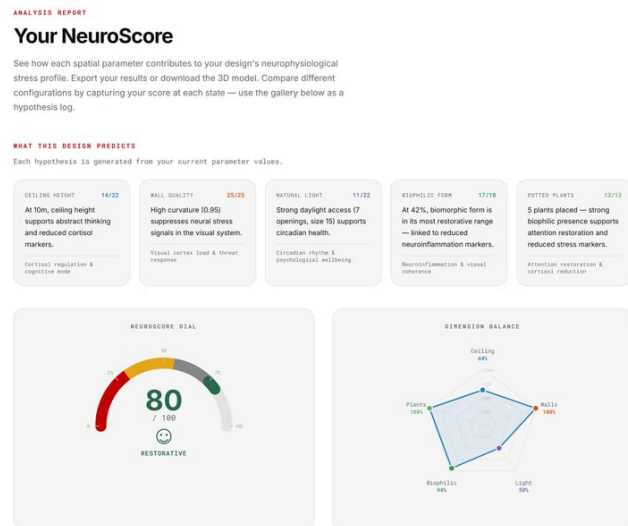


FIG 3.2 · Top of the NeuroSpace report: what this design predicts, hypothesis cards and the estimated wellbeing score dial

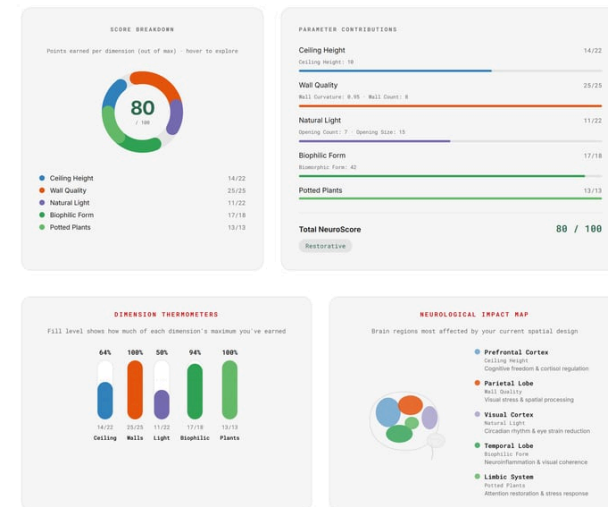


FIG 3.3 · The report's breakdown: contribution bars and a brain impact map behind the room's estimated wellbeing score



# The Lungs

*Can a dashboard run an architecture studio?*

TYPE computation & research

AWARD studio award

TEAM macad studio · data team of 3

STACK vue 3 · vite · tailwind · speckle

LINKS [live app](#) · [blog](#)

## WHAT

Santiago has an air problem, the kind that sits in the valley like a guest who will not leave. The Lungs is a hyperbuilding designed to filter 12 million m<sup>3</sup> of air a year, and the data team of three (María Sánchez Domínguez, Lakzhmy Mari Zaro and me) built the live web app the whole studio ran on while designing it.

## WHY

Three design teams, ten weeks, one shared organism: somebody had to build its nervous system, and that turned out to be the most architectural thing we designed.

## HOW

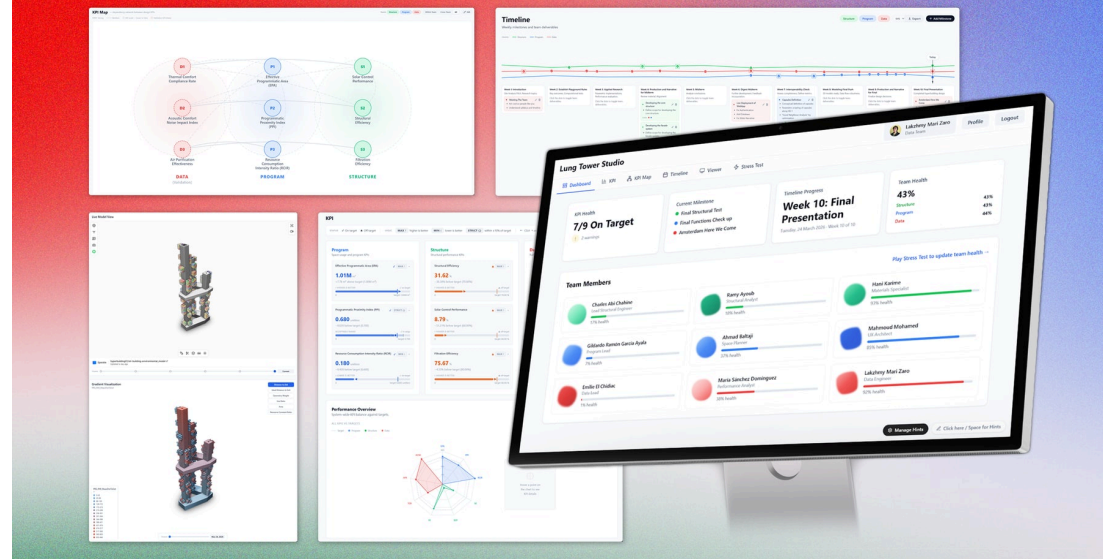
Build the platform in Vue 3, Vite and Tailwind: KPIs, team status and one timeline per team, so the overlaps show before they collide.

Stream the 3D models in through Speckle, so performance data sits on geometry, not beside it.

Keep it running while the studio designs on top of it: not documentation after the fact, the thing the studio ran on.

Score calm: a stress game the studio actually played, half telemetry (who is really using the app), half care (how everyone is holding up).

PLATE 04 · FIG 4.0



## WHAT CAME OF IT

The platform tracked three design teams across the full ten-week studio, and it is still live, with two doors: a public page where anyone can watch the tower's data, and a managerial side behind IAAC emails. Building a platform alongside the project it supports is messy, the teams' data outran the app more than once, and the mess is the lesson: the building breathes, so does the app.

PLATE 04

The Lungs

MACAD STUDIO • DATA TEAM OF 3 •  
LIVE APP

VUE 3 • VITE • TAILWIND •  
SPECKLE



FIG 4.1 • Ten weeks as a live timeline: three team tracks whose lines converge wherever the teams overlap, and the weekly deliverables the app kept honest

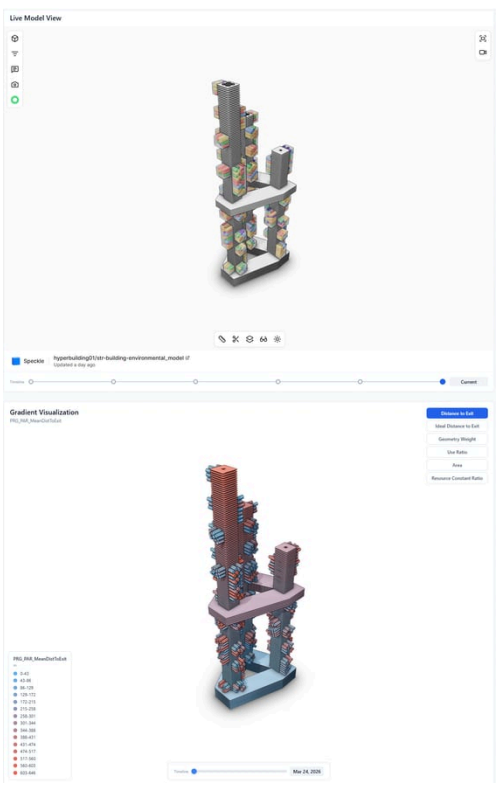


FIG 4.2 • Speckle streams the design teams' tower model into the app, so performance gradients sit on the geometry itself



FIG 4.3 • The KPI dependency network: data, program and structure indicators tied into one system the studio steered by

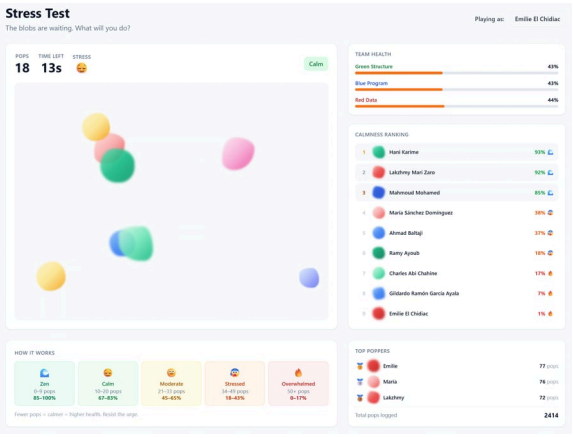


FIG 4.4 • The stress test tab, a calmness leaderboard the studio actually played; fewer pops means calmer

# The Huddle

*What if you built with the wind instead of bracing against it?*

TYPE computation & research  
AWARD studio award  
TEAM macad studio · team of 4  
STACK wasp · kangaroo · alpaca4d  
LINKS [blog](#)

## WHAT

In Punta Arenas the wind never stops, so we stopped fighting it. The Huddle is a research and education hub grown from 4×4×4 m modules aggregated along the wind itself, wrapped in an envelope of three panel types (Shields, Lenses, Gills), each answering a different face of the weather. A team of four, all hands on everything: María Sánchez Domínguez, Lakzhmy Mari Zaro, Charles Abi Chahine and me.

## WHY

Wind is the site's most abundant force, and the vernacular Kawésqar huts already knew what to do with it. Accepting the wind as a design partner, instead of a problem to brace against, is the whole project: the harshest thing on the site becomes the thing that decides where everything goes.

## HOW

WASP grows the module cluster along the wind patterns, solar exposure and program, the way the Kawésqar huts huddled.

Kangaroo settles the layout; Alpaca4D checks the structural behavior of the result.

A Global Index algorithm distributes the three envelope panels, so the facade reads the climate back to you.

Vertical turbines sit in the aggregation's own wind tunnels, harvesting the force the form was grown from.

## WHAT CAME OF IT

Wind is physics, so the planning became physics too: collision and settling dictated the groupings, and no module was placed by hand. We controlled the aggregation by tuning the forces, which is the only way it did not control us.

PLATE 05 · FIG 5.0





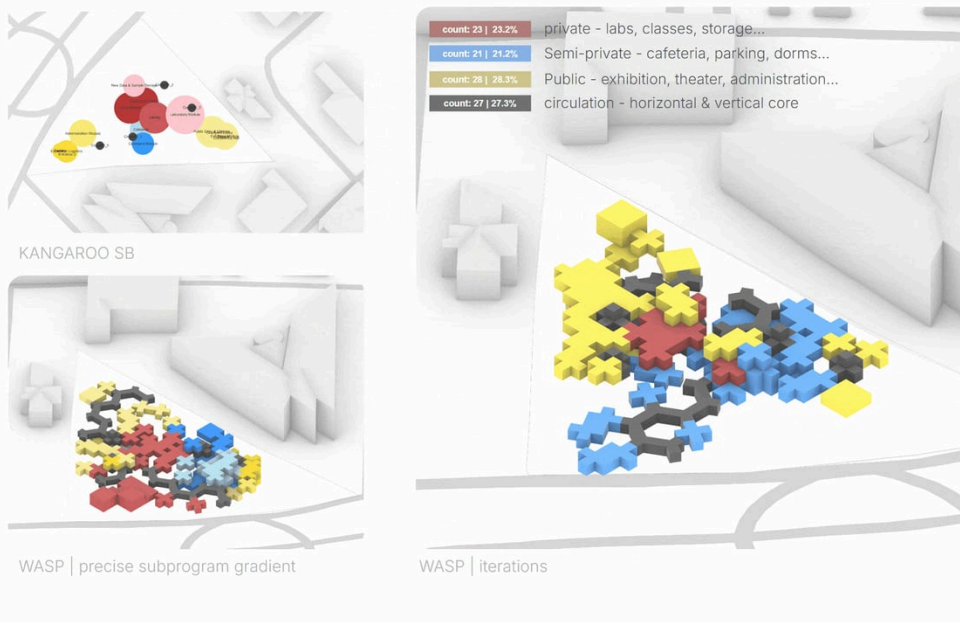


FIG 5.1 · Stills from the WASP growth study: modules aggregating along wind and program across the Punta Arenas plot

PLATE 05

The Huddle

MACAD STUDIO · TEAM OF 4

WASP · KANGAROO · ALPACA4D

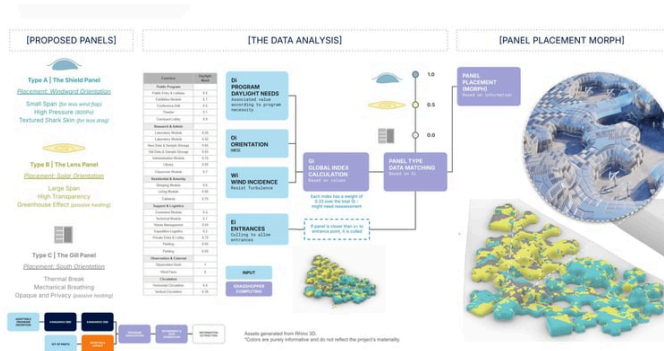


FIG 5.2 · Global Index analysis matching Shield, Lens and Gill panels to daylight, wind and program across the envelope



FIG 5.3 · Blizzard courtyard render inside The Huddle, turbines spinning between the wind-formed modules in Punta Arenas



FIG 5.4 · Street-level render of the module cluster meeting the town, the Punta Arenas sign in the foreground

# A Ballooning Market

*Can balloons roof a historic market without touching it? A pneumatic simulation.*

TYPE computation & research

TEAM macad · solo

STACK grasshopper · kangaroo · dendro · d5

LINKS [blog](#)

## WHAT

A historic Cairo market receives a new roof of pressure-packed balloons that borrows the existing steel frame without modifying it. *n.b. the frame is the client, the balloons are the tenants, the solver is the lease.* Each balloon is a Kangaroo body with collision, inflation and anchor goals; the settled cluster is meshed with Dendro and lit through CMY membranes in D5.

## WHY

A building like Bab al-Luq usually gets offered two futures: a museum piece that can never be touched, or a cold glass-and-steel box dropped in the middle and called modern. Pneumatic parasitism is the third option: a soft new life that hugs the old bones instead of replacing them.

## HOW

Trace the frame of Bab al-Luq; mark the nodes that can host anchors.

Seed balloons in the volume; declare radii, no physics yet.

Add collision and inflation goals; anchor the cluster; let the solver settle.

Mesh the settled cluster with Dendro; render daylight through the CMY membranes in D5.

## WHAT CAME OF IT

Tuning became the craft: collision too low and the balloons cuddle, too high and they panic. You tune it like a thermostat. And the run where they had no awareness of each other at all, just ghosting through one another in a chaotic, colorful mess? It stayed in the record on purpose, first in the gallery: people read honesty faster than polish.

PLATE 06 · FIG 6.0





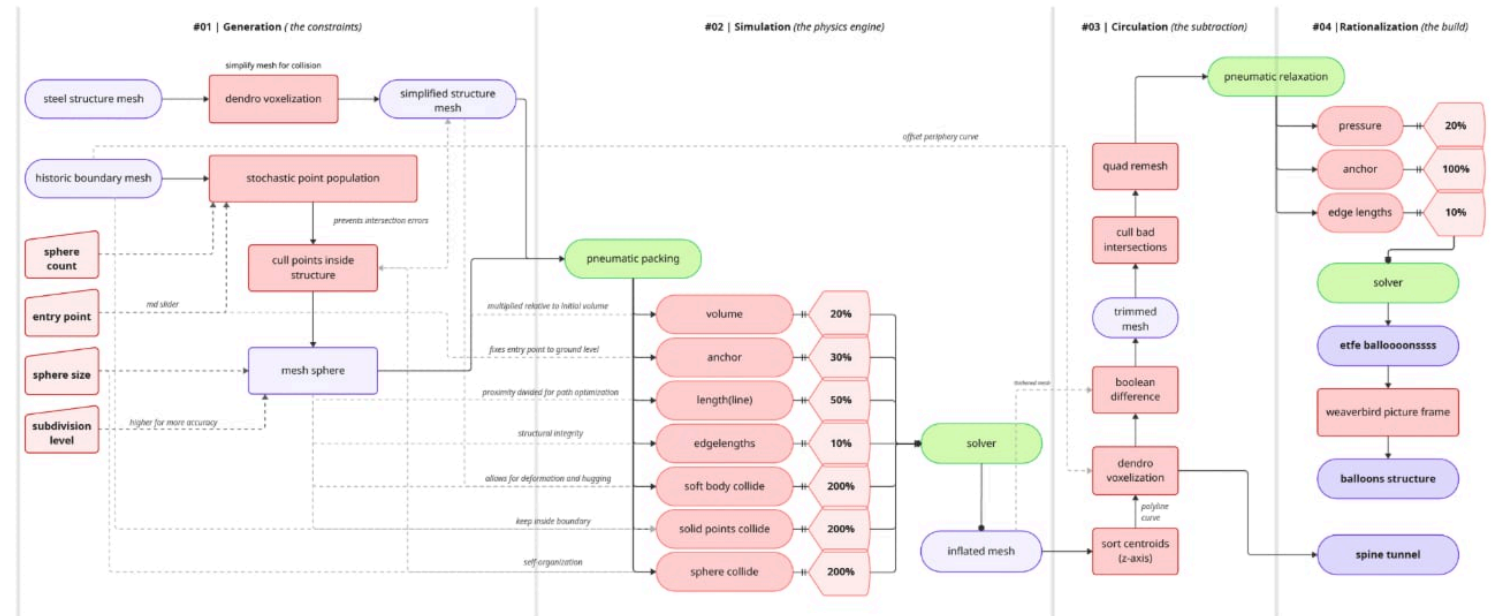


FIG 6.1 · The Grasshopper algorithm behind the inflation simulation, from constraints to Kangaroo goals to the settled cluster

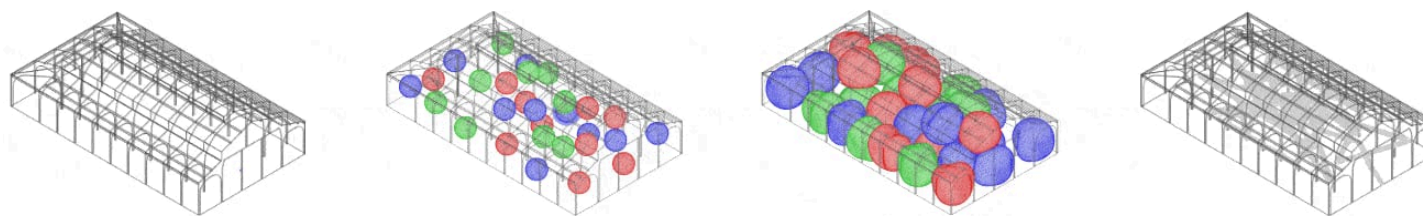


FIG 6.2 · Four stages of the Kangaroo inflation: seeding the balloons, anchoring, settling, and threading the spine path

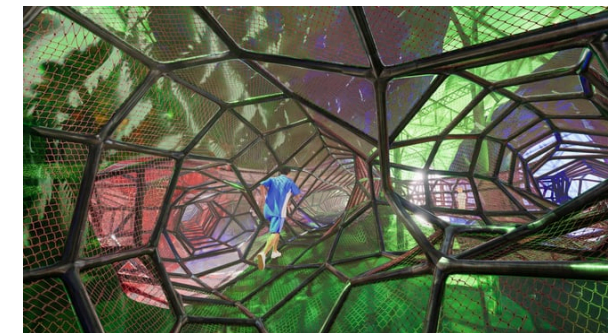


FIG 6.3 · Render of the playscape tunnel through the balloon field, a kid running the voronoi net spine

# Verve City Walk

*What survives when a parametric study meets a real drawing set?*

TYPE design & practice  
TEAM soma · design architect · 2023–24  
STACK rhino · grasshopper · revit  
LINKS [verve @ meraas](#) · [verve @ soma](#) · [district o @ soma](#) ·  
[enara @ soma](#) · [saria @ beyond](#)

## WHAT

Verve is a two-tower high-rise in Dubai: a shared amenities podium under both towers, balconies cut into the facade rather than hung off it. As a design architect at SOMA I ran the parametric facade studies in Rhino and Grasshopper, then carried what survived into the Revit model: floorplans, facade strategies, the interiors of a residential tower. The same desk carried District O, Enara and Saria from massing studies into their BIM sets.

## WHY

These are the years when I learned that a parametric study only matters if it survives contact with a drawing set. Practice is where the elegant definition meets the door schedule, and both have to win.

## HOW

Concept stays in Rhino and Grasshopper, where the massing can still change its mind: plot, podium, unit mix, all of it still cheap to move.

Design development, and everything moves into Revit: landscape planning, the typical floor plates, the interior layouts of a luxury tower. This is where gross floor area starts running the project.

Revit families built with the visualization team, and the same model going out three ways: 3D prints, renders, a VR walkthrough in Unreal.

Then every consultant back into the file, structure, MEP, facade, because the Revit model is the submission and it has to be true on the day it leaves.

## WHAT CAME OF IT

At university nobody pays for the square meters, so nobody counts them. Then you are in practice and gross floor area is the only number in the room: every balcony, every core shifted, every planted terrace, weighed against that one figure. I used to think that was the boring part. It turns out to be the part that decides what gets built. The proof of the coordination came the day a client put on a headset and walked a tower that did not exist yet.

PLATE 07 · FIG 7.0





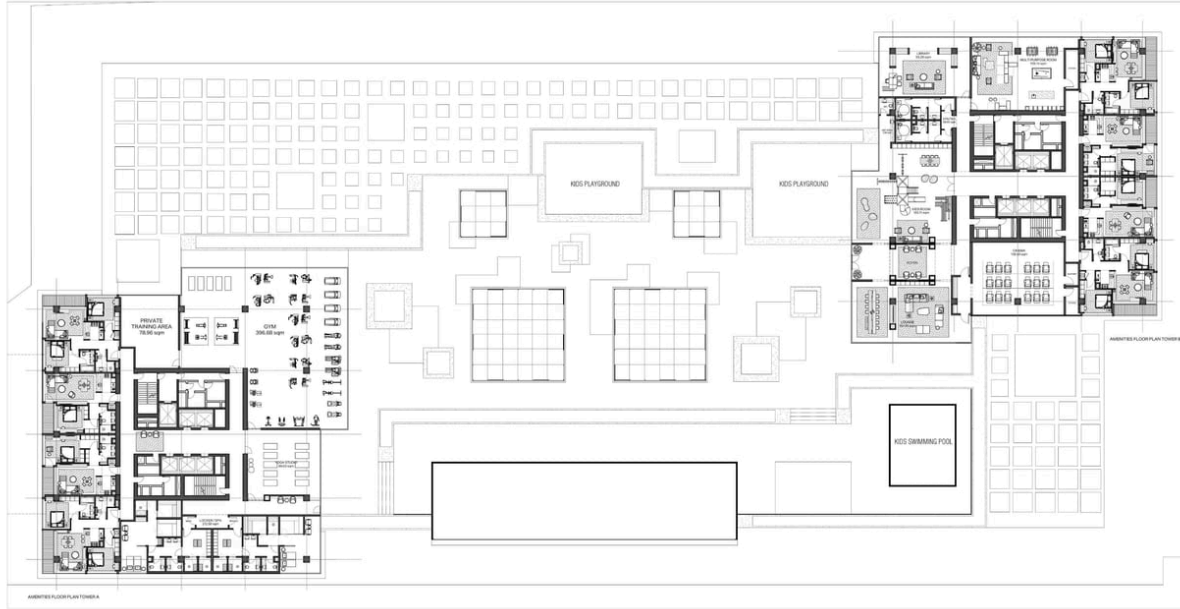


FIG 7.1 · Amenities level plan for the two Verve towers: gym, spa, yoga studio, cinema, kids' pool and playgrounds on the podium

PLATE 07

Verve City Walk

SOMA · DESIGN ARCHITECT · 2023-24

RHINO · GRASSHOPPER · REVIT



FIG 7.2 · Screen capture of the interactive 3D unit selector orbiting the twin Verve towers, one unit footprint highlighted

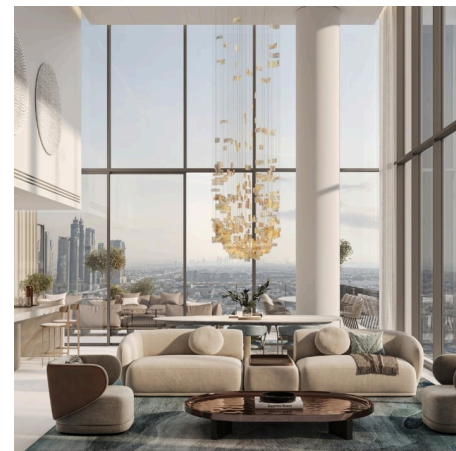
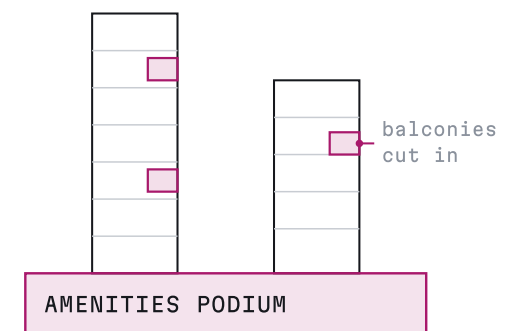


FIG 7.3 · Interior render of a double-height Verve lounge, a suspended chandelier and floor-to-ceiling glass over the skyline

TWO TOWERS, ONE PODIUM



gym · spa · yoga · cinema  
kids' pool · playgrounds

FIG 7.4 · Two towers of different height over one shared amenities podium, with the balconies cut into the facade rather than hung off it

# Falcon Square

*Can an airplane's takeoff become a monument?*

TYPE design & practice  
TEAM jemma chidiac architects · 2022  
STACK rhino · subd · archviz  
LINKS [jemma chidiac · project](#)

## WHAT

A monument for a highway roundabout in Al Khobar: MIDAN AL SAKR, the office's Moujassam Watan competition entry. Layered steel blades form a falcon, two wings of pierced Arabic calligraphy between them. As an architectural designer at Jemma Chidiac Architects, I started the concept sketches and led the design from brainstorming to proposal.

## WHY

The concept began with my ink studies of an aircraft: takeoff angles, control surfaces, the lines a plane draws when it leaves the ground. Mirrored once, the drawings opened a void; mirrored again, they closed into a falcon, the national bird carrying the national script. A monument for a roundabout has one job, to be read at speed, and a takeoff line is the fastest line there is.

## HOW

Sketch the aircraft in ink: takeoff pitches, primary and secondary control surfaces.

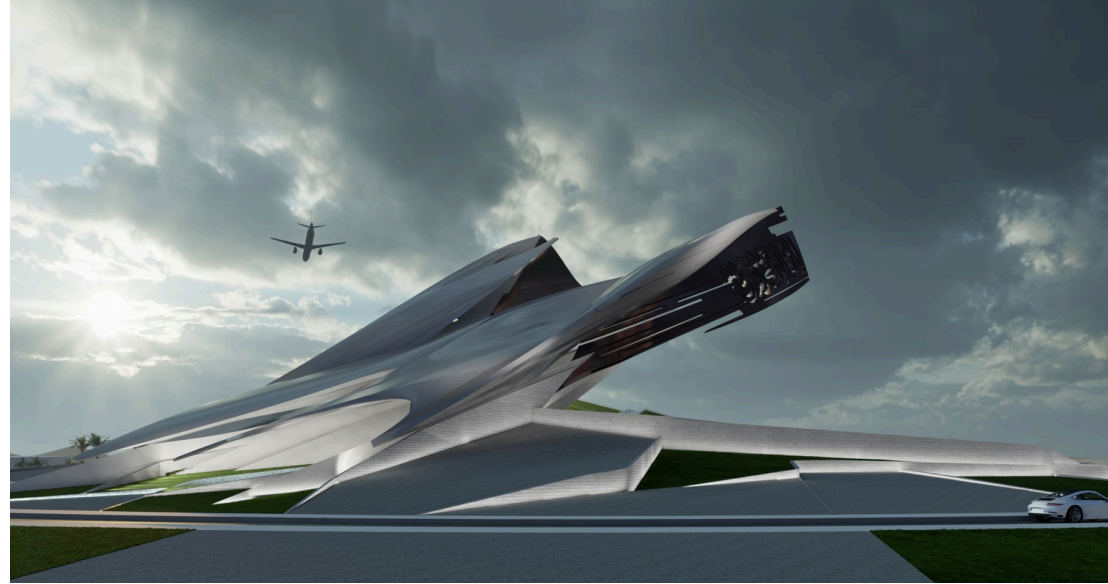
Layer the takeoff traces at different angles, then fold the set over itself until no void is read.

Build the layered blades in Rhino SubD, thread the pierced calligraphy panels between them, and re-landscape the roundabout so the ground carries the bird.

## WHAT CAME OF IT

The proposal shipped as the office's competition entry, made the finalists, and lives on the practice's site. The falcon stayed on the drawing board, which is where most competition monuments live; the sketches that started it are in the gallery.

PLATE 08 · FIG 8.0



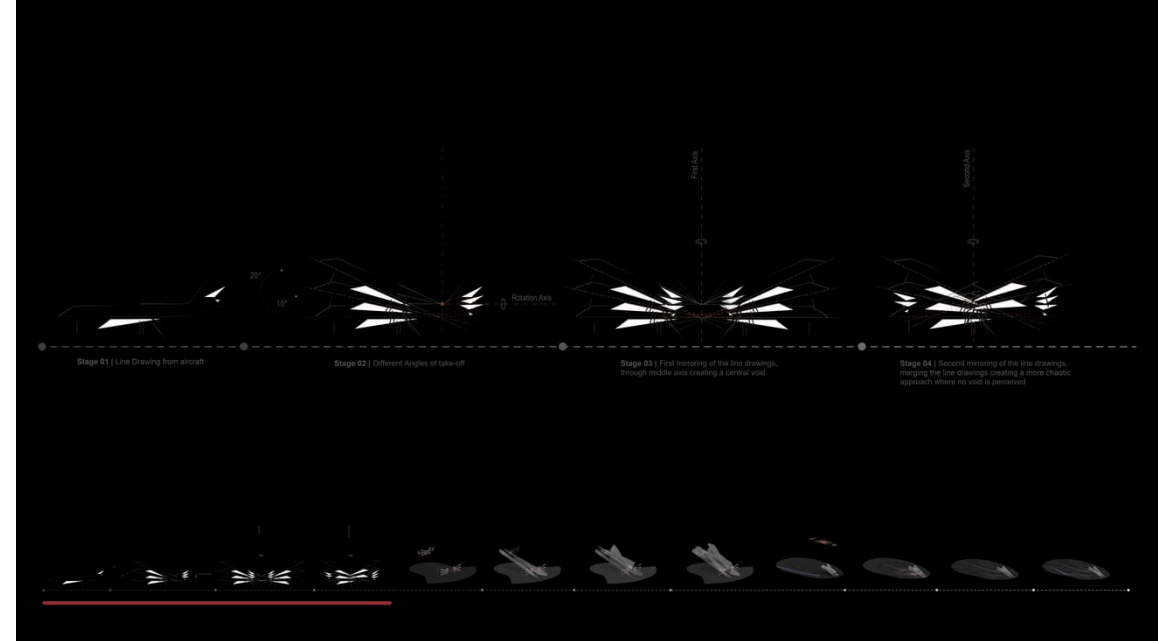


FIG 8.1 • An aircraft takeoff line becoming the falcon

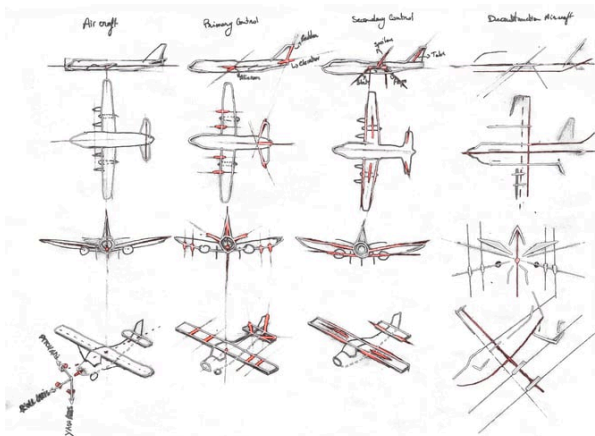


FIG 8.2 • Sketches studying aircraft anatomy: control surfaces, flaps and spoilers, and the pitch, roll and yaw axes

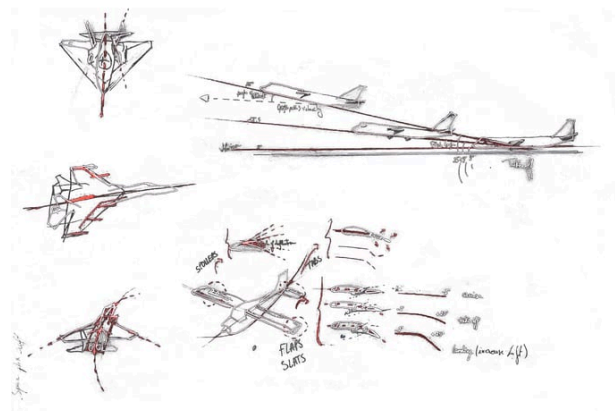


FIG 8.3 • Sketches of jets and a takeoff diagram, tracing climb angles, flaps and slats that seed the monument's lines

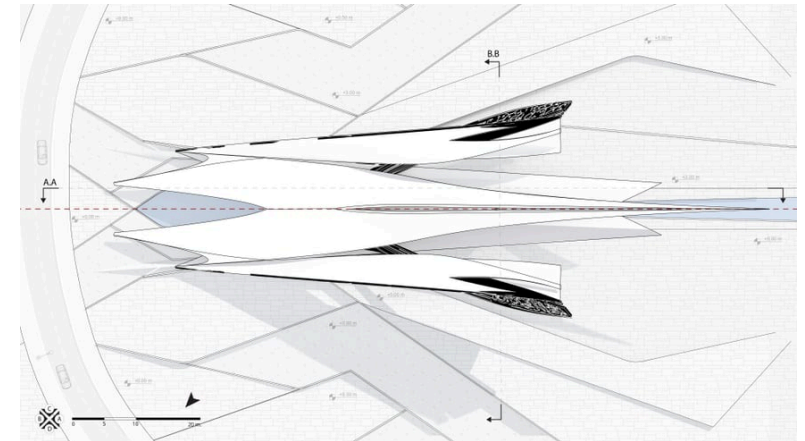


FIG 8.4 • Site plan of the falcon monument on the roundabout, mirrored wings on one axis with water channels and section lines

# Index



■ Sensi  
MACAD · +



■ IEgoarCh  
MACAD · +



■ NeuroSpace  
MACAD



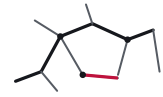
■ The Lungs  
MACAD · +



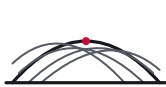
■ The Huddle  
MACAD · +



■ A Ballooning Market  
MACAD



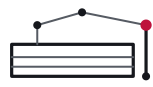
■ Encoding Urban Risk  
MACAD



▲ Cappelletti Pavilion  
MACAD



■ Optimizing for the Mind  
MACAD



■ Narkomfin as a Graph  
MACAD



■ Data into Geometry  
MACAD



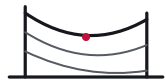
■ Tsukiji Fish Market  
MACAD



▲ Chair Simulation  
MACAD



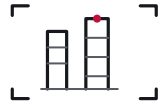
▲ Astroidal Ellipsoid  
MACAD



▲ A Playscape  
MACAD



▲ The Homage  
LAU · +



◆ Verve City Walk  
SOMA



◆ Rings of Mars  
SELF · +



◆ The Encounter  
JEMMA · +



◆ Falcon Square  
JEMMA



▲ XR for Education  
LAU

## THE LATEST THOUGHTS

|                           |          |
|---------------------------|----------|
| ○ we see in silence       | sep 2026 |
| ○ adjacency is not access | jun 2026 |
| ○ comfort as data         | may 2026 |

|                               |          |
|-------------------------------|----------|
| ○ where the plant is          | aug 2026 |
| ○ buildings that respond      | jun 2026 |
| ○ when the tool scores people | apr 2026 |

|                        |          |
|------------------------|----------|
| ○ explaining things    | jul 2026 |
| ○ experiences are data | may 2026 |
| ○ latent space         | apr 2026 |





*this is what's on my mind. what's in yours?*

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